Design from place.

Environment-aware camouflage design

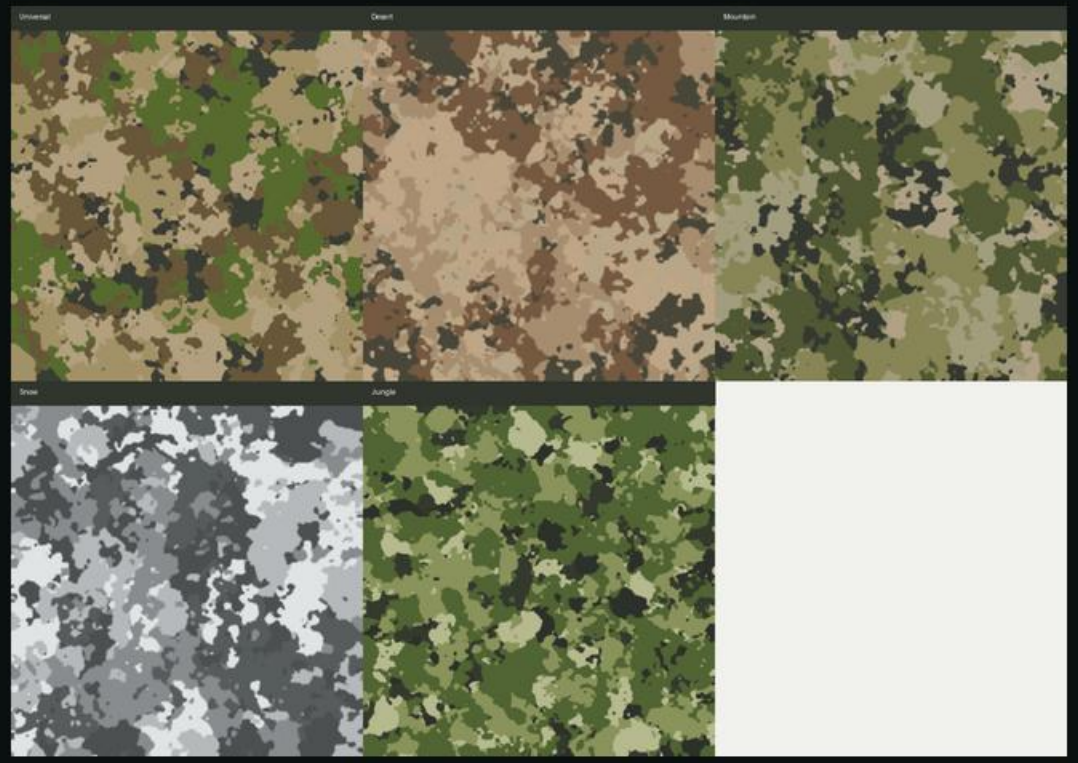


TerrainWeave / a visual research prototype

Can environments guide camouflage design?

Reference photographs -> analysis -> pattern generation -> human and computer validation -> optimization

Current baseline patterns





From pixels to palette.

A transparent, reproducible baseline

INPUT

Representative landscape photos for forest, desert, mountain, snow, urban, jungle, taiga, and tundra.

->

ANALYSIS

Dominant colors, luminance, local contrast, texture, and relevant terrain pixels.

->

OUTPUT

A 500 x 500 finite-palette pattern with irregular connected regions.

Water and sky can be excluded or down-weighted when they are not relevant to the intended terrain.

Make the pattern wearable.

A flat square is not enough

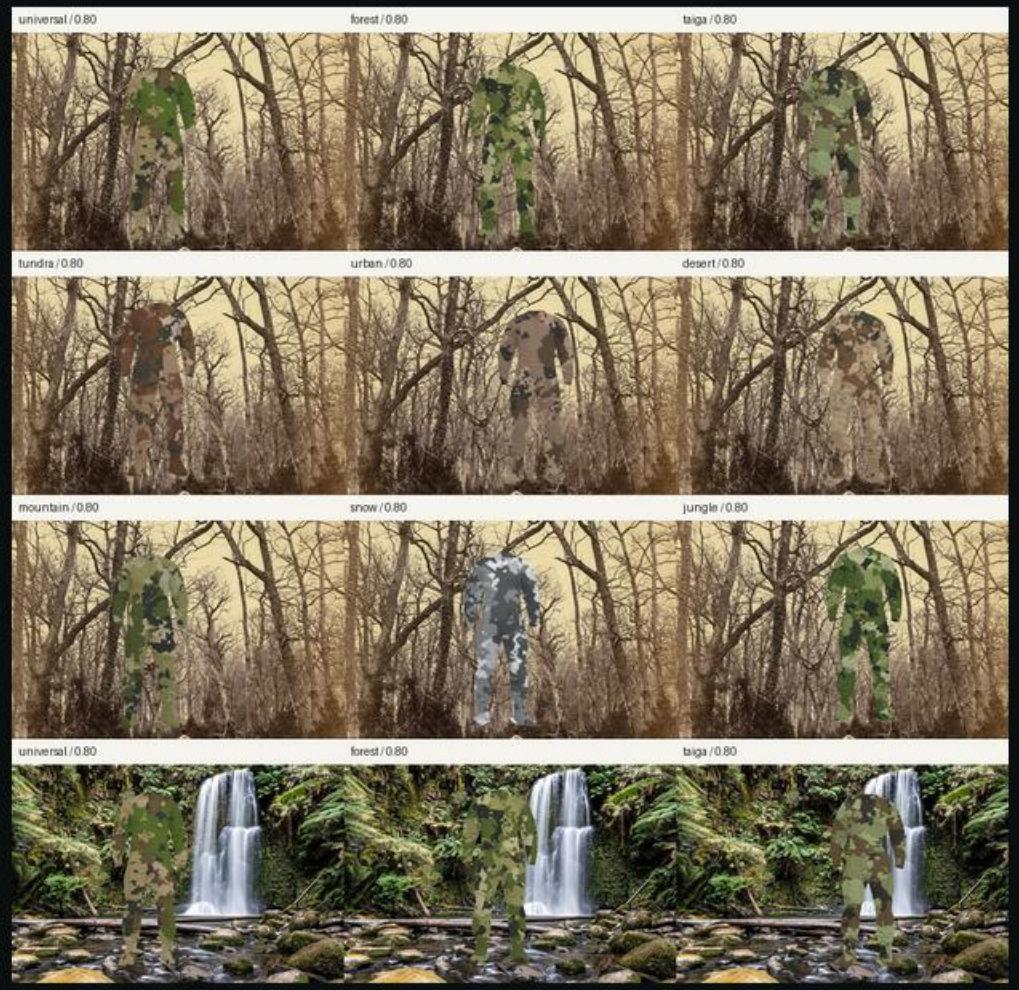


WHAT WE CHECK

- overly large patches
- excessive dark colors
- stripes or repetitive edges
- behavior at different scales

Does it disappear fast enough?

Recognition testing on real terrain



HUMAN TEST

Short exposures and randomized locations measure how quickly observers notice the figure.

COMPUTER METRICS

Color, luminance, edge, and visibility proxies compare candidate patterns.

RULE

Keep the figure on terrain, not in obvious sky or water regions.



Optimization is a loop.

Balance multiple scenes instead of fitting one photograph

A -> Generate candidates

B -> Score multiple scenes

C -> Compare human results

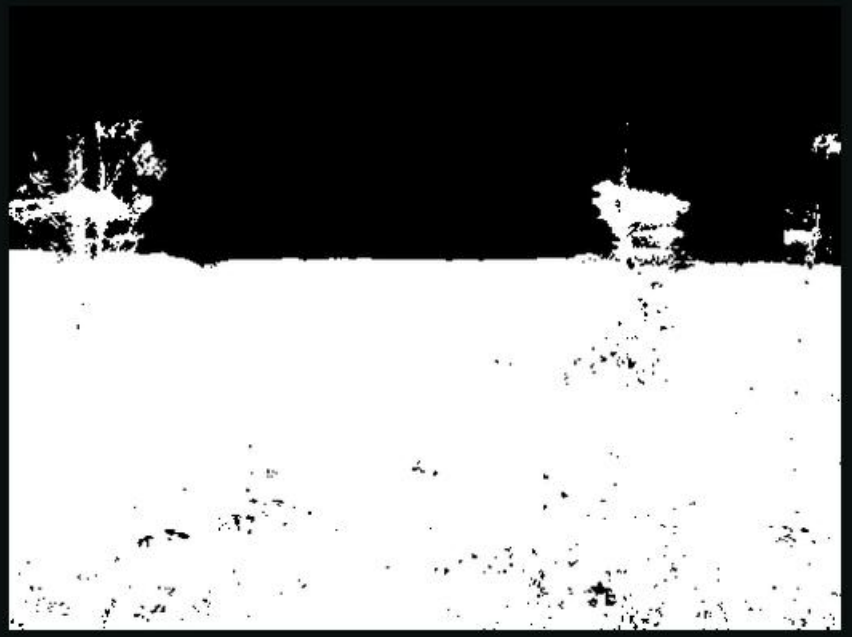
D Keep the best balanced pattern

photos -> initial pattern -> human/computer evaluation -> adjusted pattern -> repeat

Human CSV results are currently validation evidence. They are not yet used to train a neural network.

Neural input. Procedural output.

SegFormer improves scene understanding; it does not generate the camouflage



LANDSCAPE PHOTO

The pretrained model assigns semantic labels such as vegetation, terrain, sky, and water.

RELEVANT PIXELS

The pipeline can focus color and texture extraction on the terrain that matters.

LIMIT

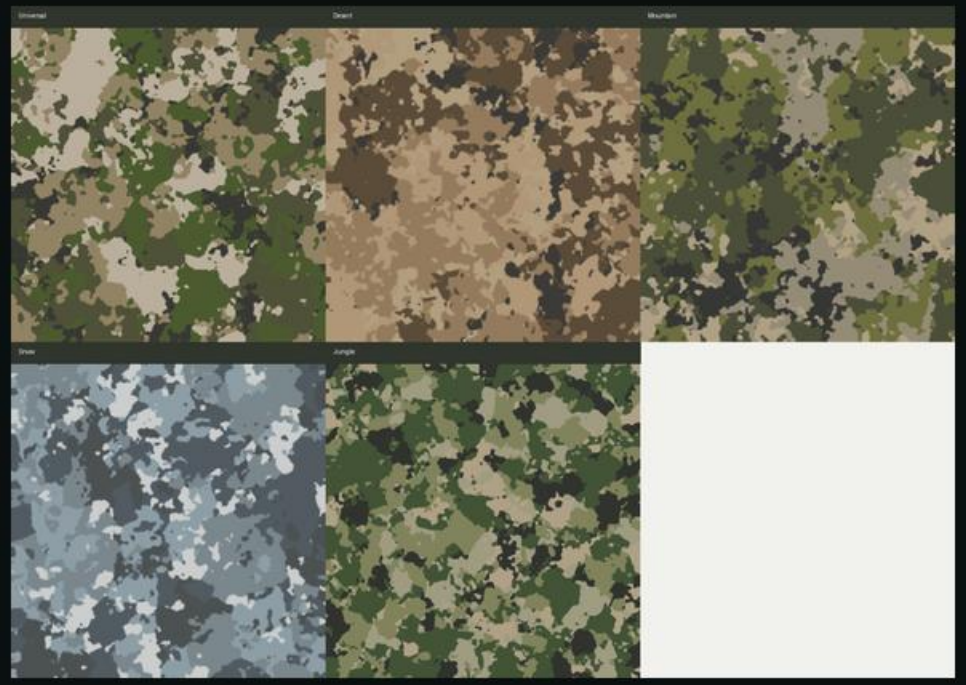
Segmentation improves the input analysis. It does not generate the camouflage or prove field performance.

From idea to testable system.

A working research prototype with a reproducible workflow

[OK] 9 environment profiles
[OK] 124 reviewed evaluation images
[OK] standard and weighted multi-objective generation
[OK] optional neural segmentation
[OK] uniform mockups and blinded human testing
[OK] JSON configurations, tests, and reports

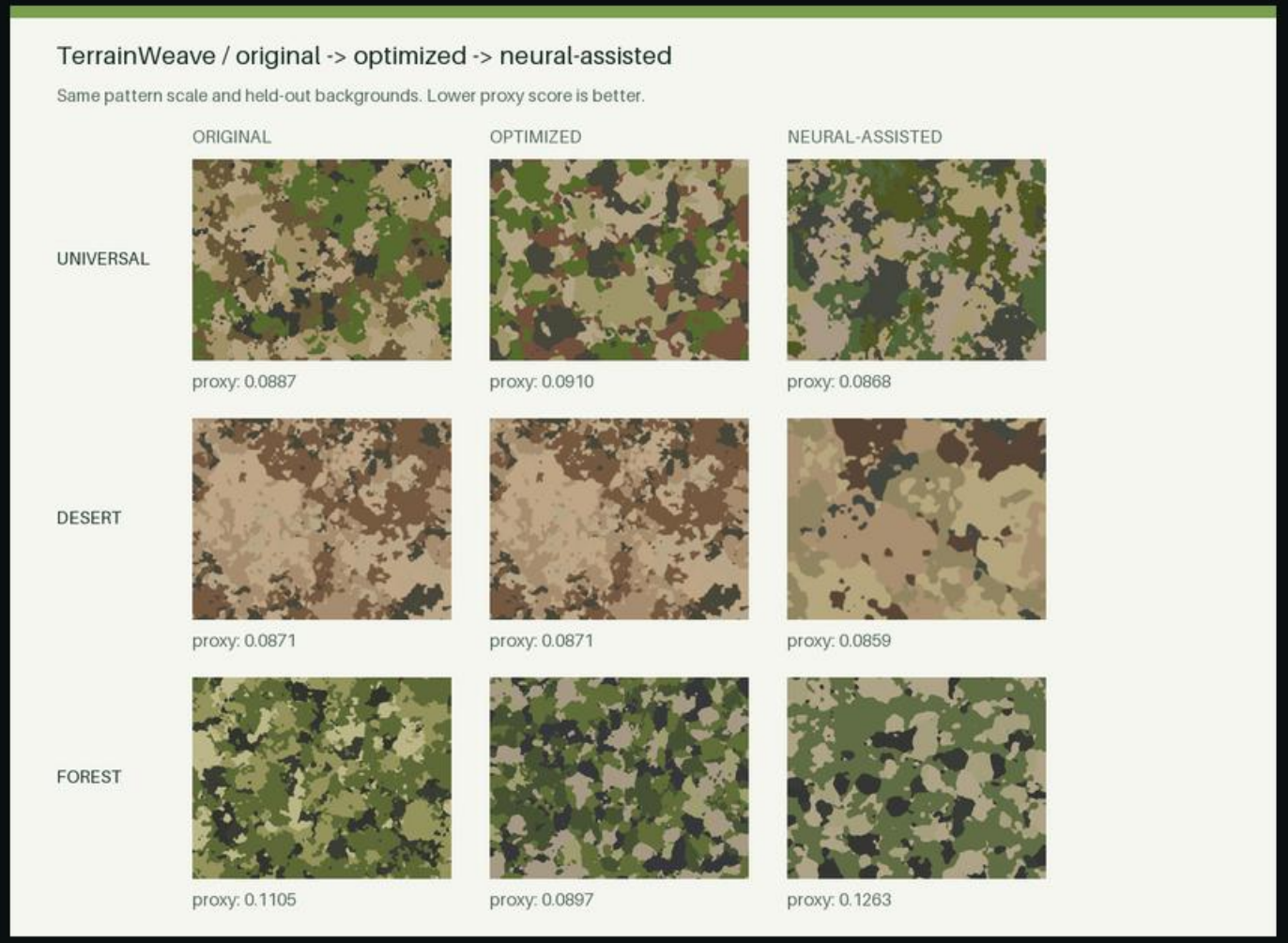
Optimized final patterns



Next milestone: learn a ranking model from a larger human-test dataset while keeping the procedural baseline.

Three versions, one forest scene.

Initial -> optimized -> neural-assisted



The comparison keeps the same pattern scale, forest background, and figure placement. Lower visibility score is preferable within this controlled experiment.